

Antiderivative

Homework

№1. Find antiderivatives of the following functions:

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| 1. $f(x) = 5;$ | 3. $f(x) = x^6;$ | 5. $f(x) = \frac{1}{x^7};$ | 7. $f(x) = \sqrt[5]{x};$ |
| 2. $f(x) = x;$ | 4. $f(x) = 2^x;$ | 6. $f(x) = \sqrt{x};$ | 8. $f(x) = x^{-5}.$ |

№2. For $f(x)$ find its antiderivative which passes through the specified point:

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| 1. $f(x) = x^3, \quad A\left(1; \frac{5}{4}\right);$ | 4. $f(x) = \frac{1}{\sin^2 x}, \quad D\left(\frac{\pi}{4}; 0\right);$ |
| 2. $f(x) = \cos x, \quad B\left(\frac{\pi}{6}; \frac{5}{2}\right);$ | 5. $f(x) = \frac{1}{\sqrt{x}}, \quad E(16; 10);$ |
| 3. $f(x) = 3^x, \quad C\left(2; \frac{9}{\ln 3}\right);$ | 6. $f(x) = \frac{1}{x}, \quad F\left(\frac{1}{e}; -2\right);$ |
| | 7. $f(x) = 2^x, \quad G(5; 1).$ |

№3. Prove that both $F_1(x) = \frac{1}{2} \sin 2x$ and $F_2(x) = -\sin^2\left(x - \frac{\pi}{4}\right)$ are antiderivatives of $f(x) = \cos 2x$. What value of C makes an equality $F_1(x) = F_2(x) + C$ true?

№4. A even function $f(x)$, defined on $\mathbb{R}$, has an antiderivative. Prove that among the antiderivatives of $\mathbb{R}$ there is an odd function.